Figure 8: Open vSwitch (Source: *http://openvswitch.org/assets/vswitch.png*)

- Standard 802.1Q VLAN model with trunk and access ports
- NIC bonding with or without LACP on upstream switch
- NetFlow, sFlow(R), and mirroring for increased visibility
- QoS (Quality of Service) configuration, plus policing
- Geneve, GRE, VXLAN, STT and LISP tunnelling
- 802.1ag connectivity fault management
- OpenFlow 1.0 plus numerous extensions
- Transactional configuration database with C and Python bindings
- High-performance forwarding using a Linux kernel module

Open vSwitch can also operate entirely in user space without assistance from a kernel module. This user space implementation should be easier to port than the kernel-based switch. OVS in user space can access Linux or DPDK devices. Note that Open vSwitch, with the user space data path and non-DPDK devices, is considered experimental and comes at the cost of performance.

Anybody who is interested can contribute here: *https://github.com/openvswitch/ovs*

The introduction of network function virtualisation (NFV) is a core structural change in the telecommunications infrastructure. NFV will bring about cost efficiencies, time-to-market improvements, new business models and services, and increased innovation to the telecommunications industry's infrastructure and applications. The evolution of NFV is real and is happening now; therefore, the telecom industry needs to consider leveraging this opportunity by heading down the migration path to NFV to start reaping the benefits.

As an emerging technology, NFV may present several challenges to network operators, such as the inability to guarantee network performance for virtual appliances, their dynamic instantiation and migration, and their efficient placement. Therefore, the challenge for network operators may be how to migrate their operations and skill base to a software based networking environment while carefully re-targeting investment to maximise the reuse of existing systems and processes. NFV may require investments in time, resources, education and operational transformation, but it may also be the most efficient strategy going forward—to keep pace with the rapid growth of the telecommunications networks of tomorrow. END

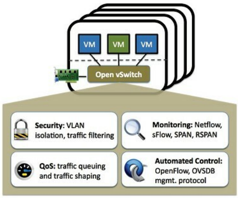

Figure 9: Open vSwitch features (Source: *http://docs.openvswitch.org/en/latest/intro/what-is-ovs/*)

References

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